

Practice Test*Equations and Inequalities*

Name _____

Date _____

Period _____

Score _____ / 21

DIRECTIONS. Show your work for every question. Write repeating decimals with a bar over the digits that repeat. Fractions do not have to be in simplest form. Where a question gives units, include them in your answer.

READ THIS FIRST. This practice test is the test — same twelve questions, same order, same point values, only the numbers are different. Nothing on the test will look like anything you have not seen here.

	What it covers	Questions	Points
A	Equivalent forms of rational numbers	1, 2	7
B	Writing and solving an equation from a situation	3, 7	4
C	Solving two-step equations	4, 5, 6	3
D	Deciding whether a value solves an inequality	8	1
E	Solving and graphing inequalities	9, 10, 11	3
F	An inequality in a real-world situation	12	3

1. Complete the table by writing the equivalent value in each empty cell.

3 points

	Fraction	Decimal	Percent
a.	5/9		
b.		0.65	
c.			48%

2. Devon opens a bag of fruit chews with five flavors. The table shows what part of the bag is each flavor.

4 points

Flavor	Amount
Cherry	25%
Grape	7/45
Lemon	0.15
Orange	$33.\overline{3}\%$
Berry	$0.\overline{1}$

- a.** What percentage of the bag is Lemon? *1 point*

- b.** Rewrite the amount that is Grape as a decimal.

1 point

Percentage	
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Decimal	
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c. What fraction of the bag is Berry? *1 point*

d. Rewrite the amount that is Cherry as a decimal. *1 point*

Fraction	
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Decimal	
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3. Rashad is saving for a summer camp. He is paid the same amount every week. Each week he uses \$415 of his pay for expenses and saves everything that is left. After 9 weeks he has saved \$423. *2 points*

a. Write an equation that can be used to determine the amount, x , in dollars that Rashad is paid each week. *1 point*

Equation	
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b. Solve the equation. *1 point*

Solution	
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Solve each equation. Show your work.

4. $-6(w + 5) = 96$
1 point

5. $n \div (-4) + 2.6 = -5.9$
1 point

6. $2\frac{5}{6} = 7\frac{1}{4} - k$
1 point

Solution	
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Solution	
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Solution	
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7. A server at the Lakeside Diner earns \$9.40 per hour plus \$64.60 in tips.

2 points

- a. Write an equation that can be used to determine the number of hours, h , the server worked to earn \$135.10.

1 point

Equation	
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- b. Solve the equation.

1 point

Solution	
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8. Circle every value below that is a solution to $-2x + 5 \geq 17$.

1 point

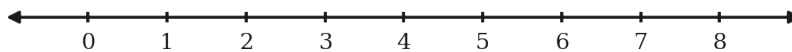
-11 $-6\frac{1}{3}$ -6 -5 0 4

Solve each inequality. Show your work, then graph the solution set on the number line.

9. $-9.7 > -2x - 1.3$

1 point

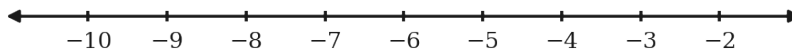
Solution	
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10. $\frac{2}{3}(18 - 15x) > 64$

1 point

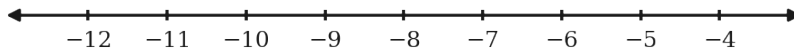
Solution	
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11. $-6(4x + 7) \geq 150$

1 point

Solution	
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12. The chess club is renting a van for a tournament. The rental company charges a flat fee of \$180 plus \$16 for each student who rides. The club has raised \$700 and wants to take as many students as possible.

3 points

- a. Write an inequality that can be used to determine the number of students, s , who can ride the van.

1 point

Inequality	
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- b. Solve the inequality.

1 point

Solution	
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- c. How many students can ride? Justify your answer mathematically.

1 point